# Self-Evolving Recommender Systems: A Survey and an Evaluation Audit

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## Abstract

In 2026 a new kind of recommender research appeared: large language model (LLM) agents that propose changes to a recommender's configuration or source code, train the changed model, and keep what the evaluation says is better. Within nine months about thirty such systems were reported, from open-source research harnesses to production loops at Google, Meta, Amazon, Alibaba, Kuaishou, Tencent and NetEase. We organise them, together with search-space AutoML and the agentic recommenders that update their own memory or weights, by what the loop changes, who proposes the change, and which chain of signals decides whether it survives, and we synthesise their recurring designs and the changes they keep. Because a self-evolving system is only as good as the signal it selects on, we then audit how 27 configuration-, code- and component-level loops are evaluated, coding each paper's full text against eight criteria with released evidence. Of the 17 loops that select candidates offline, three of the ten that report ranking metrics rank the full catalogue, seven do not say which data keep or reject candidates, four select on the data they report, and three state that the test set was used once; none reports its headline result with the spread over repeated evolution sessions, and three compare against a non-LLM search at equal budget. Of the 19 loops that run online experiments, none corrects for the many rounds it tests and one confirms its final candidate in a fresh experiment. Where transfer to held-out data is checked, some selected gains survive and some do not. Four loops report that their agent inflated the metric or leaked the target; where the paper says what caught it, it was people or the agent's own diagnosis, not the acceptance rule. We distil these findings into a two-tier reporting standard mapped to the audit criteria.

## Keywords

recommender systems, LLM agents, AutoML, program evolution, evaluation, reproducibility

## 1 Introduction

Recommender models have long been improved by a loop that engineers run by hand: propose a change, train, compare offline, and launch the winners into an online experiment. Automating parts of this loop is an old goal. Neural architecture and embedding search, mostly for click-through-rate (CTR) prediction models [89, 137, 187, 192], and loss search [79] automated the choice inside a space that a person had written down [194]. What changed in 2026 is that the proposer became an LLM agent that writes code. Following FunSearch [130], AlphaEvolve [113] and open implementations such as OpenEvolve [114] and ShinkaEvolve [70], agents now edit the model, loss and training code of recommenders directly. Google reported such a loop for YouTube ranking [147] and Self-EvolveRec released an open harness [66] in February 2026; within months, open harnesses such as RecHarness [87] and systems at Meta [76, 104], Alibaba [51, 106], Kuaishou [71, 167, 201], Tencent [5, 94], NetEase [185] and others followed.

These systems are optimisers, and an optimiser exploits whatever its objective rewards. A loop that evaluates hundreds of agent-written candidates on one validation set and keeps the maximum is the setting in which the known weaknesses of recommender evaluation—sampled metrics [69], temporal leakage [58], untuned baselines [27, 127], and the reuse of a holdout for adaptive decisions [13, 31]—are most likely to turn into reported gains. The agent also controls the code that produces its score. Five of the systems we read report that their agent won on the metric without a real improvement. For example, RecEvolve's agent inflated its proxy metrics by shrinking the batch size, and with it the number of in-batch negatives [117], and an AI co-scientist for search ranking produced implementations that "improved $\mathcal{M}$ while being wrong (e.g., silently increasing model size)" [159]. Figure 1 shows the loop these systems share and the points where the evidence for a gain can fail.

Existing surveys do not cover this development. Surveys of LLM agents for recommendation treat the agent as the recommender, a conversational interface or a user simulator [54, 120, 190]; surveys of self-evolving agents treat the agent's own prompts, memory, tools or weights as the object that evolves, and mention recommendation at most in passing (a two-sentence mention of self-optimised fine-tuning and a recommendation task in a benchmark it describes in [4], none in [33]); the most recent roadmap for agentic recommender systems mentions the "agent as engineer" direction only in passing, in two sentences that cite AgentX and the Google loop [84]; and the most recent peer-reviewed survey of agentic recommenders organises 47 studies by where the agent sits in the serving workflow, not by what changes the recommender [164]. A general survey of how automated research is evaluated covers no recommender system [183]. Closest to our audit, a concurrent controlled study shows on CTR and learning-to-rank data that exact validation feedback lets a coding agent overfit the validation split, and that a feedback contract with a quarantined test set removes the spurious gain on a task with no signal [171]. Beel et al. call for autonomous recommender-systems research labs and for benchmarks to evaluate them [10]. RMD-Bench [119] fixes protocols for cross-dataset and cross-scale discovery and ExecML-HSTU [23] measures whether agents' edits execute correctly, but no shared benchmark yet compares evolvers under a sealed test, full-ranking evaluation and equal-budget null controls.

This survey makes four contributions.

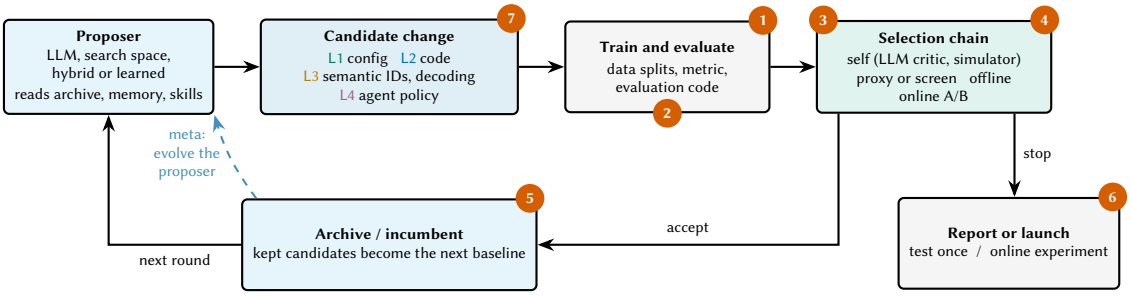



**Figure 1: Anatomy of a self-evolving recommender. A proposer changes some part of the recommender (the evolved object); the changed system is trained and evaluated; a chain of selection signals decides what is kept and becomes the next baseline, and the loop may also change its own proposer (meta). Red markers show where the evidence for a reported gain can fail, with the audit finding and criterion that check it: 1 the agent edits evaluation code, reads test data or games the metric (F7, C9); 2 sampled or unstated candidate sets (F1, C1); 3 selection on unstated or reported data, without a single test evaluation (F2, F3; C2, C3); 4 a proxy or self-generated signal that does not transfer (F5, C5); 5 a gain inside run-to-run noise becomes the new baseline (F4, C4); 6 online rounds selected without correction or fresh confirmation (F9, R9); 7 gains measured against an untuned starting point, with no equal-budget null search (F6, F8; C6, C8).**

(1) **A taxonomy** (§4) that separates what the loop changes (the evolved object), who proposes the change (an enumerated search space, an LLM, or a learned proposer, possibly itself evolved) and which chain of signals decides whether it survives.
(2) **A synthesis of the systems** (§5): their recurring designs, what their loops actually kept, and what is open source.
(3) **An evaluation audit** (§6) of the 27 configuration-, code- and component-level recommender loops on eight scored criteria, coded from the full text with verbatim evidence that we release, with offline-selecting and online-selecting loops reported separately, and a taxonomy of the shortcuts agents found.
(4) **A reporting standard** (§7) with a minimum and a recommended tier, mapped to the audit criteria, together with open problems (§8).

## 2 Scope and Survey Method

*Definition.* We call a recommender system *self-evolving* when an automated proposer changes some part of it, the changed system is evaluated, and the evaluation decides which changes are kept, repeatedly and without a human choosing each change. Humans may gate, veto or approve machine-generated candidates, review a loop at checkpoints, or roll back a change; a system is out of scope when a human, not an evaluation, makes the keep-or-reject decision. This includes systems whose proposer is a search algorithm over a fixed space (background, §3), but our focus is on LLM proposers.

*Inclusion.* We include (i) systems that change a recommender's or ranker's configuration, features, code or components under an evaluation loop; (ii) agentic or LLM recommenders whose policy, memory, skills, prompts or weights improve from feedback they generate; and (iii) benchmarks, environments and surveys for these. We exclude LLM recommenders without a self-improvement loop, conversational recommendation in general, general hyperparameter-optimisation libraries, methods that evolve the recommendations rather than the recommender (such as QD-Rec's quality–diversity archive of items and hypotheses [56]), and continual-learning methods that call themselves self-evolving but have no proposer or selection step, such as LION's sparse key–value memory for a generative recommender [85]. Seven studies outside recommendation are included as comparison points because they inform evaluation: learning to rank [118], search ranking [159], e-commerce search strategy [150], lexical retrieval [112], generative document retrieval [111], lifetime-value (LTV) prediction [156], and a controlled study of agent feedback on public CTR and learning-to-rank (LTR) data [171].

*Search protocol.* The cut-off date is 22 October 2026. We started from keyword searches over arXiv, OpenAlex and Semantic Scholar ("self-evolving recommender", "LLM agent recommender model optimisation", "agentic harness recommender", "LLM-driven architecture search CTR" and survey queries), then walked the reference lists of the 2026 systems (RecHarness, EvoRec, AgentX-Model, RecEvolve) and of the agentic-recommender roadmap backwards and their titles forwards, and finished with about thirty full-text arXiv phrasings combined with company names. Generic keyword search returned mostly unrelated "agent for $X$" papers; the citation graph of the 2026 systems was the productive source. This first sweep yielded 22 of the audited systems and five comparison points. A second sweep then ran 190 further arXiv queries (817 new candidates, all screened by title), enumerated 13 ACM proceedings volumes through OpenAlex (RecSys, KDD, SIGIR and WWW 2025–2026 including the WWW companions and the February KDD 2026 cycle, CIKM 2025 and WSDM 2026; 6,874 records screened by title), followed forward citations of 16 seed systems in Semantic Scholar, and searched industry engineering blogs. It found nine further candidates: five entered the audit as core systems, two as comparison points, and two are known only from their abstracts (§6). CIKM 2026, DBLP and OpenReview could not be searched automatically. Every arXiv entry in our bibliography was generated from the arXiv API record, and every code repository we list was checked to exist.

**Table 1: Audit codebook. C7 is recorded as free text and discussed in F8; the other eight criteria are scored (values in Table 4).**

| | |
|---|---|
| C1 | Metric: full-catalogue ranking, sampled ranking (with $N$ negatives), AUC/loss, or online only |
| C2 | Selection signal: which split or signal accepts or rejects each candidate |
| C3 | Test discipline: is the selected candidate evaluated on a held-out test once, after search? |
| C4 | Variance: repeated seeds or repeated evolution runs for headline numbers |
| C5 | Overfitting check: validation-to-test gap or held-out transfer of the selected candidate |
| C6 | Null control: random search, HPO or another non-LLM search at equal budget |
| C7 | Budget: what is counted (iterations, GPU time, tokens, money) |
| C8 | Baselines: are starting points and compared models tuned? |
| C9 | Isolation: can the agent reach test data or evaluation code? Reported incidents of gaming or leakage are recorded separately. |

*Audit procedure.* For each configuration- or code-level system, we read the full text (arXiv HTML or PDF, including appendices) and coded nine criteria (Table 1). A criterion is coded "not stated" when the paper does not say, which is different from "no". Each coded cell is backed by a verbatim quote of at most 25 words with a section pointer; the data file and the evidence are released[1] so that the coding can be checked and corrected. Extraction and verification were LLM-assisted: LLM agents (Claude) drafted the evidence for each cell from the full text, two further agent passes re-checked every coded cell and every cited sentence against its source, and the authors reviewed the coding and each correction against the paper text. For systems whose launch decision is an online experiment, we read the offline criteria as describing the offline screen, and assess the launch evidence separately (R9 in §7).

## 3 Background

*Search-space AutoML for recommenders.* AutoML for recommenders searched for feature-interaction structures (AutoCTR [137], AutoFIS [89]), embedding dimensions (AutoEmb [192], AutoDim [193]), whole architectures with weight sharing (NASRec [187], DNS-Rec [186]) and loss functions (AutoLossGen [79]); Zheng et al. survey this line [194]. Each method can only return a design that its space contains. Evaluation is mostly validation logloss or AUC on public CTR data, or standard rating and sequential benchmarks, with standard splits. Even in this setting, careful re-evaluation of architecture search found that the search phase often added little over random sampling and that the evaluation protocol dominated reported differences [75, 166, 173]. The community answered with tabular and surrogate benchmarks that make equal-budget comparisons cheap [30, 32, 172] and with a best-practice checklist [86]. The LLM loops below inherit these problems with a larger, open-ended space and with control over the evaluation code.

[1] https://github.com/shangliy/recforge/tree/master/papers/self-evolving-recsys-survey/audit

*LLM program evolution and ML-engineering agents.* FunSearch [130] and AlphaEvolve [113] pair an LLM that proposes code with an automatic evaluator and an archive of programs. Open systems followed: OpenEvolve [114], ShinkaEvolve [70] and DeepEvolve [92]. ML-engineering agents—AIDE [60], MLE-STAR [107], R&D-Agent [170], AIRA [144], DS-Agent [42], AutoML-Agent [145], Agent Laboratory [132] and the AI Scientist [97, 165]—apply the same pattern to whole training pipelines, and recent work optimises the agent harness itself [73, 96, 155, 163]. Their benchmarks contain at most one recommendation task: MLE-bench includes the H&M fashion-recommendation competition [17] and MLRC-Bench a next-product-recommendation task [189]. MLRC-Bench shows agents only the development split and scores the best development snapshot once on test, whereas MLGym's `validate` command returns test-set metrics during the session [109].

*Generative recommenders.* HSTU [177] and TIGER [122] recast recommendation as sequence generation over item IDs or semantic IDs, and OneRec unifies retrieval and ranking in one generative model that is further aligned by iterative preference optimisation on its own samples [28, 196, 197]. These models add new objects that a loop can evolve: tokenisers, codebooks, decoding procedures and generative losses.

*Evaluation methodology.* Tuned simple baselines beat many published neural recommenders [27, 127]; a loss change alone can reorder SASRec [61] and BERT4Rec [67, 139], and BERT4Rec's reported advantage depends on much longer training [121]; sampled metrics are inconsistent with full-ranking metrics and can reverse model rankings [48, 69, 125]; leave-one-out splits that ignore the global timeline leak future information [48, 58], and the split choice changes rankings [103]. Systematic re-examinations found that most neural recommenders could not be reproduced or were beaten by tuned simple baselines [26], that evaluation choices varied widely across papers [141], and that with untuned baselines any proposed model can win [133]. Libraries and open benchmarks such as RecBole [191], Elliot [3], DaisyRec [140], BARS [198] and FuxiCTR [199] standardise protocols for human-designed models. For adaptive reuse of a holdout, theory predicts overfitting [13, 31]; empirically, a meta-analysis of 120 Kaggle competitions found little [129]. For LLM loops specifically, keep-if-better acceptance is selection under noise, with selection-set scores well above held-out accuracy when the selection set is small [52]; best-of-$N$ runs produce apparent gains from selection alone [41]; for LLM research agents, the scarcity of overfitting has been tied to how compressible their successful strategies are [12]; and AI-scientist systems hide post-hoc selection, leakage and metric misuse [98]. Beyond recommendation, results depend on seeds and on the budget spent tuning them [14, 29, 47], leakage is a recurring cause of irreproducible ML-based science [63], and agent benchmarks are sensitive to cost, budget and task-setup flaws [55, 64, 154, 200]. We return to this tension in §6. Figure 2 places these precursors next to the 2026 systems.

## 4 A Taxonomy

A recent blueprint for self-evolving agentic recommenders distinguishes reinforcement-learning evolution in defined action spaces

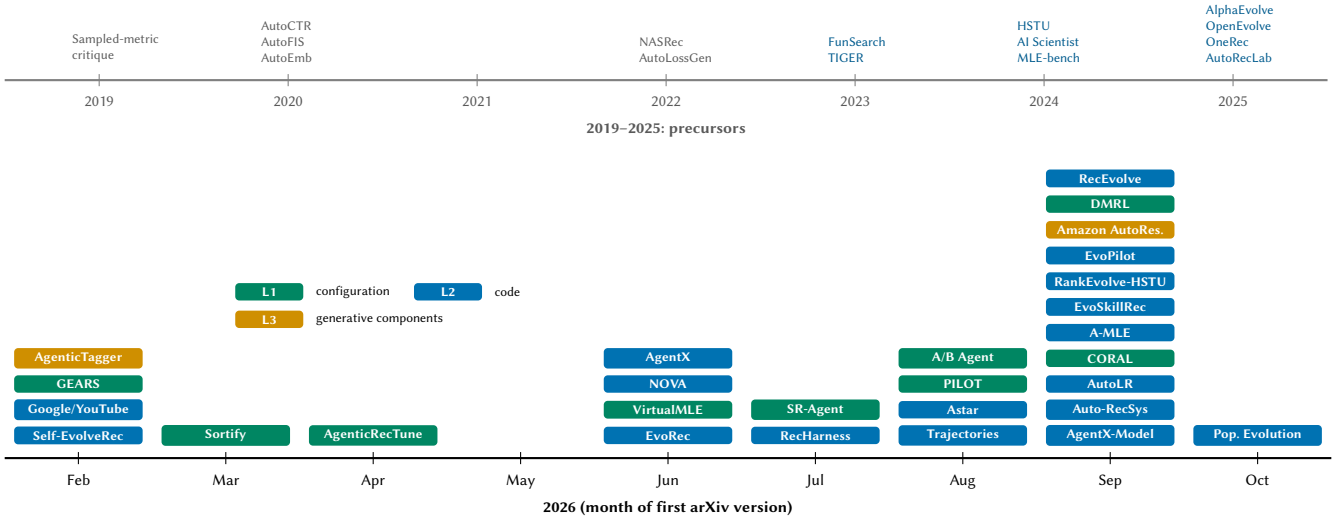



**Figure 2: From search-space AutoML to agent-driven evolution. Top: precursors by year (grey: search-space AutoML and evaluation critique; blue: program evolution, ML-engineering agents and generative recommenders). Bottom: the 2026 configuration-, code- and component-level systems surveyed here, by month of their first arXiv version; the colour gives the highest level of the evolved object. AgenticTagger is surveyed but not audited; comparison points outside recommendation and L4 systems are omitted.**

from LLM generation and selection in open design spaces [50]. We describe a loop by three independent choices: what it changes, who proposes the change, and which signals decide whether the change survives (Table 2). Separating the proposer from the object resolves a confusion in earlier framings, where "AutoML" named a kind of proposer (search over a written-down space) but was listed as if it were a kind of object.

*Axis 1: the evolved object.* **L1 configuration**: values in an existing configuration—hyperparameters, fusion weights, retrieval budgets, thresholds, strategy parameters. **L2 code**: executable model, loss, training or feature code, however small the edit. **L3 generative components**: semantic-ID construction, item tokenisers and vocabularies, and decoding of a generative recommender; L3 takes precedence when such a component is changed through configuration or code. **L4 agent policy**: the prompts, memory, skills or weights of a recommender that is itself an LLM agent. A loop that changes several objects carries several labels (L1+L2 when an agent edits both learning rates and model code); 10 of the 27 audited loops do.

*Axis 2: the proposer.* An *enumerated space*: classical search (grid, Bayesian optimisation, evolutionary or differentiable architecture search) over options a person wrote down; this is search-space AutoML. An *LLM*: a language model writes the change, free-form or inside human-defined edit arms. A *hybrid*: a classical selector chooses the direction and an LLM writes the change (RecHarness's Thompson sampling over edit arms [87]), or an LLM steers a classical search (Sortify's controller around Optuna [25]). A *learned proposer*: a model trained on the system's own history proposes the next direction (Astar [51]). Orthogonal to these, a loop is *meta-evolving* when it changes the proposer's own instructions, tools, skills or weights and reuses them later; keeping an archive or a log of past results does not count. By this rule 13 of the 27 audited loops are meta-evolving (Table 3).

*Axis 3: the selection chain.* A candidate passes an ordered chain of signals, one of which decides inside the loop. **Self**: an LLM critic, council, auditor or simulator judges the candidate before or instead of measurement. **Proxy**: a reduced-scale screen, a proxy model, a training-loss curve or a replay. **Offline**: evaluation of the trained candidate on held-out data. **Online**: an A/B experiment. Industrial loops typically chain several, for example self→offline→online; whether the offline split is a validation or a test split, and whether it is stated, is a property of the evaluation that we audit (§6), not of the design.

Each signal fails in a characteristic way, and the failures are of two kinds [99]. *Regressional* failures come from selecting the maximum of noisy estimates: an offline split reused over hundreds of candidates rewards noise [31], a proxy ranks candidates differently from the full model, an online round selected as best overstates its effect [72], and an LLM judge from the same model family as the proposer shares its errors. *Adversarial* failures come from an optimiser that changes what is measured: an agent that can edit the evaluation code, the batch size of an in-batch metric or the evaluation date can raise the score without improving the recommender [135, 195]. The audit checks the first kind with C2–C6 and the second with C9.

# 5 Systems

## 5.1 Configuration-level loops (L1)

L1 systems let an LLM change numbers or switches while the model code stays fixed. VirtualMLE tunes SASRec and HSTU through

**Table 2: Taxonomy: evolved object (rows) × proposer (columns). Multi-object loops appear in the row of their highest level. † open-source code verified; m meta-evolving (the loop also changes its proposer); u not yet audited. Selection chains of the audited systems are in Table 3.**

| | Enumerated space | LLM | Hybrid (search + LLM) | Learned proposer |
|---|---|---|---|---|
| L1 configuration | AutoEmb [192], AutoDim [193] | VirtualMLE†m [16], CORAL [6], AgenticRecTunem [160], SR-Agent [167], A/B Agent [59]; *comparison:* PEAR [150] | Sortifym [25], GEARS [175], PILOTm [82] | DMRLm [188] |
| L2 code | AutoCTR† [137], AutoFIS† [89], NASRec† [187], DNS-Rec [186], AutoLossGen† [79] | Self-EvolveRec†m [66], Google/YouTube [147], EvoRecm [106], NOVAm [94], AgentXm [71], AgentX-Modelm [169], Auto-RecSysm [76], RecEvolve [117], A-MLE [39], RankEvolve-HSTU [23], Population Evolution [119], Trajectories-to-Evidence [201], EvoPilot [35], Meta REA (blog) [104], AutoModelu [180], RuleAgentu [153]; *comparison:* EvoRank† [118], AI Co-Scientist [159], RankEvolve-IR† [112], AgentLTV† [156], Feedback Contract [171] | RecHarness† [87], AutoLR [185], EvoSkillRec†m [78] | Astarm [51] |
| L3 components | | Amazon AutoResearch [18], AgenticTaggeru [161]; *comparison:* Evolving Semantic Indexes [111] | | |
| L4 agent policy | Selected mostly on self-generated feedback. Memory and reflection: AgentCF† [181], AgentCF++† [93], SAGER [142], RecNet [74], AtomRec [53], MemRec† [21], Re2LLM [151], RecPilot [115]; weights: OneRec [28], SPRec† [36], Rec-R1† [83], CoARS [152], RSIR† [182], EvoReason [202]; prompts and tool descriptions: RecPrompt [90], Spotify [116] | | | |

an execute–reflect loop with a hierarchical memory that transfers across datasets [16]. In production, CORAL allocates retrieval budgets and serving treatments [6], AgenticRecTune tunes fusion weights and thresholds across pre-ranking, ranking and re-ranking [160], Sortify searches the exchange rates of a sorting formula with Optuna [2] while an LLM meta-controller recalibrates the search's offline-to-online transfer intercepts and penalties [25], GEARS selects segment-level treatment policies [175], and SR-Agent edits post-ranking diversity and exposure strategies through four bounded action types [167]. Taobao's PILOT has LLM roles choose ranking-objective weights for user segments from a legal-command envelope generated by deterministic services [82]; Kuaishou's A/B Agent iterates strategy parameters across online rounds with a hierarchical experience tree [59]; and DMRL trains a small LLM with reinforcement learning to rewrite the skill document that an ad-tuning agent follows [188]. LLM-driven hyperparameter optimisation [95, 184] is the general-ML form of L1 and is routinely compared against Bayesian optimisation or random search, the null control that few recommender loops run (F6). Most L1 systems in production select directly on online experiments; an e-commerce search loop adds a multi-fidelity ladder from shadow-traffic proxies to decision-grade online tests [150].

## 5.2 Code-level loops in the open (L2)

**Self-EvolveRec** [66] runs an AlphaEvolve-style loop over the whole pipeline code of NCF [46], NGCF [149], SASRec and MoRec [174] seeds; besides validation metrics it receives directional critiques from an LLM user simulator and from a diagnosis tool that evolves with the model. **RecHarness** [87] routes each round with Thompson sampling over human-defined "edit arms" (schedules, dropout, depth, pooling, features, loss) plus "jump" arms for structural changes, uses validation feedback only, evaluates the incumbent on test once, and caps compute at 43,200 GPU-seconds; HSTU is one of its five backbones. **EvoSkillRec** [78] evolves architectures as typed "skill genomes" of executable modules and promotes validated modules into a skill library. **RankEvolve-HSTU** [23] composes Claude Code and Codex under an executable operating protocol in which the two agents review and repair each other's edits to HSTU, and introduces ExecML-HSTU, a benchmark of how often agent edits are executed correctly. **Population Evolution** [119] runs several evolution searches that share evidence across related datasets, screens candidates on peer datasets (and, for LLM and VLM targets, on small training subsets), and promotes survivors to larger scales; it introduces RMD-Bench. AutoRecLab [9, 10] generates recommender experiments from natural-language descriptions; it automates experimentation rather than evolving a model, and we do not audit it. Two loops evolve data and features rather than the model: RuleAgent has a language agent propose rules for denoising implicit feedback, retrains with an unlearning shortcut, and keeps rules by validation performance [153], and Agent0 tunes the prompts that turn raw text into categorical features through feedback from a downstream oracle [136].

## 5.3 Code-level loops in industry (L2)

Most industrial systems share a two-stage design: an offline loop screens agent-written candidates, and online experiments decide launches. Google's loop for YouTube ranking [147] has Gemini personas propose optimiser, architecture and reward changes, promotes on validation loss (with a proxy metric for reward changes) and then on north-star A/B metrics, and reports LLM costs of about $20,000 over six months. RecEvolve [117] runs the full research loop on a production two-tower retrieval model. Meta's REA plans, launches and debugs ads-ranking experiments with up-front GPU budget approval [104]; A-MLE explores a portfolio of ads-ranking models with human checkpoints [39]; Auto-RecSys runs idea- and execution-evolution loops over multi-day training jobs [76]; and RankEvolve-HSTU and Population Evolution above also come from Meta. Alibaba's EvoRec [106] pairs research and code agents with

a skill evolver, and Astar [51] trains an 8B model on the commit history of an advertising system to propose the next evolution direction; from Astar's authors, AutoModel organises an industrial recommender as evolution agents for models, features and online experimentation, with a case study of reproducing published methods and comparing them offline against production [180]. AgentX, deployed at Kuaishou [71], iterates online strategies and model research, and its successor AgentX-Model [169] runs research lineages (reproduce, follow up, compose, diagnose) over production rankers, and a Kuaishou paper converts research-agent trajectories into auditable evidence [201]. Tencent's NOVA [94] adds verification gates that reject recommender-invalid edits such as target leakage, NetEase's AutoLR [185] reviews proposals with a multi-expert proposal council under a limited trial budget, and Tencent's RecSys Factory bounds agent autonomy to typed decision points in fixed pipelines [5]. A Trip.com co-scientist for travel search ranking [159], LLM evolution of learning-to-rank pipelines (EvoRank [118]) and of lexical retrieval functions [112] are close relatives outside recommendation. Meta's EvoPilot runs a 37-day campaign of agent-proposed configuration and retraining changes to video-retrieval models behind a fail-closed verifier and catalogues the failures it met [35], and Amazon's AutoResearch rewrites the training code of a book-retrieval model and tunes a semantic-ID tokenizer in a twelve-week campaign of more than 220 experiments, naming the failure modes it met [18]. Adjacent loops evolve ads kernels [80], off-policy-evaluation code [158], search-relevance models [146] and an LLM policy layer trained against a learned evaluator [179], and an operations agent for Kuaishou's search engine evolves its skills from on-call engineers' corrections [88]. Feature-engineering loops are close relatives: an error-driven loop turns AUC ranking inversions into feedback for an LLM feature generator [8], a planner-guided agent writes feature code for recommendation models with human checkpoints [143], and LATTEArena compares LLM feature-engineering methods on cost as well as accuracy [43].

## 5.4 Components of generative recommenders (L3)

Generative recommenders expose objects that do not exist in ID-based models. Evolving Semantic Indexes, a generative document-retrieval system that we treat as a comparison point, runs LLM-driven program evolution over semantic-ID construction, collision handling, decoding and the generator's training objective, using a one-million-parameter proxy decoder as fitness [111]. AgenticTagger has an LLM refine a hierarchical item-descriptor vocabulary used as tokens [161], and a generative recommender can reassign its own item tokenisation during training [20]. Amazon's AutoResearch is the first audited production loop that edits a semantic-ID tokenizer, and the first in which the agent gamed a tokenizer metric [18]. Feedback-grounded policy discovery distils LLM-discovered natural-language policies into a semantic-ID generator [24]. Evolving Semantic Indexes evolves TIGER-style semantic-ID components for document retrieval but releases no code; no open system yet does so for recommendation under a shared protocol.

## 5.5 Self-improving agentic recommenders (L4)

When the recommender is itself an LLM agent, the evolving object is its memory, skills, prompts or weights, and the feedback is mostly generated by the system. Memory and reflection methods update user and item agents (AgentCF [181], AgentCF++ [93]), per-user policy skills (SAGER [142]), propagation modules optimised with textual gradients (RecNet [74]), and structured memories that are rewritten as interactions arrive (AtomRec [53], AMEM4Rec [110], STEAM [81], MARS [134], MemRec [21]), together with reflection-built hint bases (Re2LLM [151]) and per-user rubrics refined from user feedback (RecPilot [115]), and recommendation and user agents that refine each other's inferences over rounds without amplifying popularity or position bias [15]; CoVeMem instead replaces rewritten text memory with frozen collaborative-filtering states learned by gradient training [19]. Weight-level self-improvement includes OneRec's alignment on its own samples, first by iterative DPO [28] and then by on-policy RL [196, 197], self-play debiasing (SPRec [36]), RL against a recommender's metric (Rec-R1 [83]), co-evolving recommender and user agents (CoARS [152]), reasoning primitives and a teacher updated from a student's on-policy outcomes (EvoReason [202]), and recursive self-training on filtered synthetic sequences [45, 182], where progress can also sit in earlier generations, not only in the latest model [162]. RecPrompt iteratively rewrites a news recommender's prompt with an optimiser [90], and Spotify improves a conversational recommendation agent's prompts, few-shot examples and tool descriptions with a coding agent, evaluated by an LLM judge on synthetic conversations [116]. Simulated environments and benchmarks for such agents include RecoWorld [91], the user simulators Agent4Rec [178] and RecAgent [148], and the verifiable $\tau$-Rec benchmark [108]. A controlled pilot found multi-agent recommendation pipelines not uniformly better than a single-shot baseline [101], a reminder that L4 gains also need controlled baselines.

## 5.6 Evolving the evolver (meta)

Thirteen of the 27 audited loops change their own proposer. AgentX's SGPO applies semantic-gradient updates to its agents' instructions and validation rules [71]; EvoRec, AgenticRecTune and EvoSkill-Rec grow skill libraries from experiment outcomes [78, 106, 160]; Auto-RecSys rewrites execution playbooks [76]; VirtualMLE and PILOT carry tuning rules or methodology across datasets and experiments [16, 82]; Astar retrains its proposer on each iteration's verified outcomes [51]; and DMRL trains a small LLM with reinforcement learning to rewrite the skill document that an ad-tuning agent follows [188]. These are the recommender counterparts of harness optimisation [73, 96, 163]. Outside recommendation, AIDE$^2$ keeps changes to its own code that perform best on hidden evaluations and shows that they transfer to four held-out benchmarks [138]; separating selection tasks from held-out tasks is what the meta-evolving recommenders do not yet report. Meta-evolution inherits the selection problem one level up: the harness is kept or discarded by a score of its own. In a pre-registered study of LLM research-ideation harnesses by one of us, outside recommendation and not yet peer reviewed, a harness optimised against an LLM judge raised the judge's scores while the share of its hypotheses that held under

measurement fell, on two hypothesis spaces with measured ground truth and for two Claude models (Sonnet and Opus) [168].

## 5.7 Recurring designs

Table 3 places the 27 audited loops on the three axes and adds the role of humans; the released `design.csv` also records each loop's memory, acceptance rule, guards and LLM, with evidence. Four patterns recur.

*Where the chain ends.* Eight loops end at an offline evaluation: three open harnesses (Self-EvolveRec, VirtualMLE, EvoSkillRec) and five industrial research loops whose papers report offline results, or in Auto-RecSys's case only operational ones (A-MLE, Auto-RecSys, RankEvolve-HSTU, Population Evolution, Amazon AutoResearch). The other 19 pass candidates to an online experiment, and 14 of them involve humans: eleven approve launches, rounds or candidates, and three check or intervene. The two ends are exposed to different failures. A loop that ends offline controls the code and data that produce its final number, so the adversarial failures matter most; a loop that ends online cannot change the measurement easily but selects on expensive, noisy experiments, so the regressional ones do.

*LLM review before measurement.* Eight loops make an LLM critic, council or auditor a gate that every candidate must pass before it is measured (Table 3, chains starting with S); others, such as RecEvolve, A-MLE and RankEvolve-HSTU, use LLM critics to rank or filter ideas. These gates are cheap and catch invalid edits—NOVA's verifier rejects target leakage [94]—but they add a selection stage whose judge often comes from the same model family as the proposer, the condition under which a self-generated signal is least reliable [40, 168].

*Hybrids that keep a classical selector.* Six loops leave the choice of direction or the numeric search to a classical method and use the LLM to write or steer: Thompson sampling over edit arms (RecHarness), an evidence-weighted selector (AutoLR), typed operators over a skill library (EvoSkillRec), Optuna (Sortify), Pareto enumeration (GEARS) and a deterministic action enumerator (PILOT). They are the easiest designs to compare against an equal-budget null, because the classical part already defines the space.

*Humans in the loop.* Only nine papers describe no human role in the loop. Five have a person approve or review each round or candidate (AgentX, AgenticRecTune, A-MLE, EvoPilot, A/B Agent); this does not remove them from scope, because the keep-or-reject decision is still made by an evaluation, but it means that reported gains are those of a human–agent system.

## 5.8 What the loops kept

Three papers report no kept edit (Auto-RecSys, AutoLR, DMRL), Astar reports only proposals in case studies, and A-MLE only technique families; the other 22 name the changes their loops kept or launched, and we coded these 87 edits by kind (`design.csv`; the tally is generated by `summarize.py`). The eight configuration-only loops kept strategy parameters (17 of their 20 edits): fusion weights, sorting-formula coefficients, retrieval budgets, thresholds. The fourteen code-level loops kept most often architectural changes (29 of 67) and loss or objective changes (12), then optimiser and schedule settings (8), data and negative sampling (6), and a few strategy, regularisation, capacity and feature changes. Code-level loops are therefore not merely expensive hyperparameter optimisers, which matters for the choice of null control (R5 in §7): an equal-budget HPO run is a necessary but not a sufficient null for them.

Many of the named edits are established techniques transplanted into a particular model: InfoNCE losses and hard negatives [18, 66, 106], DCN-V2 layers and cosine learning-rate decay [117], SwiGLU feed-forward blocks [16, 94], gated paths [147], RMSNorm [201], expert pools [169] and checkpoint averaging [119]. This is useful engineering, and finding which known technique helps a given production model is much of what human model iteration does. It also suggests a null that no paper runs: a scripted sweep over a catalogue of known techniques at the same budget. One caveat applies: papers report the edits they find interesting, not all kept edits.

# 6 How Are Self-Evolving Recommenders Evaluated? An Audit

*Audited set.* We audited the 27 recommendation loops at levels L1–L3 that keep or reject candidates by evaluation, and the seven comparison points of §2. Because offline hygiene criteria mean little for a loop whose keep-or-reject decision is an online experiment, we report two strata: 17 *offline-selecting* loops, which decide on offline or proxy evaluations (some later confirm online), and 9 *online-selecting* loops, which decide on A/B experiments and are assessed on the design of those experiments; Auto-RecSys reports only operational outcomes and makes no accuracy claim. Meta's REA is described only in a blog post and is not audited, and two NeurIPS 2026 workshop papers are known to us only from their abstracts: MLE Agents, which kept 325 of 1,097 agent-written changes to production search rankers [77], and an open-ended research loop on the public KuaiRand [38] and KuaiRec [37] data [22]; we will code them once the full texts are available. Four borderline systems were read but not counted: RecSys Factory [5] (an operations platform in which an operator's approval, not an evaluation, keeps or rejects each proposed override), AutoRecLab [9] (experiment generation), AURA [65] (stops at pull requests) and Spotify's prompt loop [116] (an L4 system). Feedback-grounded policy discovery [24] evolves per-user LLM-teacher text, an L4-like object, and AgenticTagger [161], RuleAgent [153], Agent0 [136] and AutoModel (a case study) [180], found late, have not yet been coded. Table 4 gives the coding and Figure 3 summarises it; every count below is generated from the released data file. Where a count rests on cells coded "not stated", we give the range it would span if every such cell were the favourable answer.

*F1: Candidate sets are rarely full, and often unstated.* Ten offline-selecting loops report ranking metrics. Three rank the full catalogue (RankEvolve-HSTU, Population Evolution, and the retrieval system of Amazon's AutoResearch); two pair each test item with 99 sampled negatives (Self-EvolveRec, RecHarness); five do not say what the candidate set is, so between three and eight of the ten use full ranking. For RecEvolve, the fact that shrinking the batch inflated its metrics (F7) suggests in-batch negatives. In Amazon's loop the agent designed a text-only fallback that "expanded catalog coverage by 5.8×" [18], which by our reading changed the candidate set during the campaign. Six offline-selecting loops evaluate ranking-stage

**Table 3: Design of the 27 audited loops. Chain: S self (LLM critic, council or simulator), P proxy or screen, O offline evaluation, A online experiment; the stage in bold decides inside the loop. Meta: the loop changes its own proposer. Humans: approve each round or candidate (each), approve launches (launch), check at stage boundaries or intervene (checkpoints), or none stated. Google's offline stage keeps candidates and its online stage chooses launches; Sortify's proxy search proposes each round and the rounds are chosen online; DMRL's deciding proxy is a long-term predictor applied to online experiment outcomes, so the audit counts it as online-selecting. Evidence for every cell is in the released `design-evidence.md`.**

| System | Object | Proposer | Chain | Meta | Humans | System | Object | Proposer | Chain | Meta | Humans |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Self-EvolveRec | L2 | LLM | **O** | Y | none | Sortify | L1 | hybrid | P**A**A | Y | launch |
| RecHarness | L1+L2 | hybrid | **O**A | N | none | GEARS | L1 | hybrid | **A** (logged) | N | launch |
| Google/YouTube | L1+L2 | LLM | **O**A | N | launch | RecEvolve | L2 | LLM | P**O**A | N | checkpoints |
| EvoRec | L1+L2 | LLM | **O**A | Y | none | Astar | L2 | learned | P**O**A | Y | none |
| VirtualMLE | L1 | LLM | **O** | Y | none | A-MLE | L1+L2 | LLM | **O** | N | each |
| NOVA | L1+L2 | LLM | S**O**A | Y | checkpoints | SR-Agent | L1 | LLM | SP**A** | N | launch |
| AgentX | L1+L2 | LLM | S/O**A** | Y | each | EvoSkillRec | L2 | hybrid | **O** | Y | none |
| AgentX-Model | L2 | LLM | S**O**A | Y | launch | RankEvolve-HSTU | L1+L2 | LLM | **O** | N | checkpoints |
| Auto-RecSys | L2 | LLM | **O** | Y | checkpoints | Population Evol. | L2 | LLM | **OPO** | N | none |
| AutoLR | L1+L2 | hybrid | S**O**A | N | launch | Trajectories | L2 | LLM | S**O**A | N | none |
| CORAL | L1 | LLM | **A** | N | checkpoints | EvoPilot | L1+L2 | LLM | **O**A | N | each |
| AgenticRecTune | L1 | LLM | S**A** | Y | each | Amazon AutoRes. | L1–L3 | LLM | **O** | N | checkpoints |
| | | | | | | PILOT | L1 | hybrid | P**A**A | Y | launch |
| | | | | | | A/B Agent | L1 | LLM | S**A** | N | each |
| | | | | | | DMRL | L1 | learned | A**P** | Y | none |

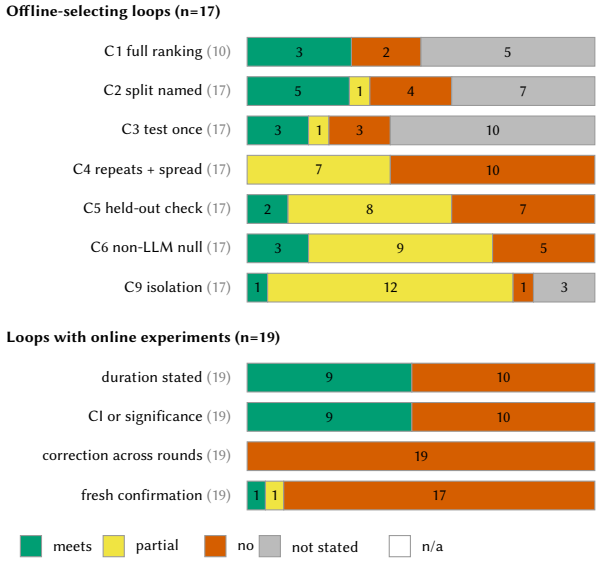



**Figure 3: Audit summary. Top: the 17 offline-selecting loops (C1 among the 10 that report ranking metrics; the others report AUC or a loss). Bottom: the design of the online experiments reported by 19 loops. C9 counts the isolation design; incidents of gaming or leakage are reported separately (F7). Generated from the released `audit.csv`.**

CTR models with AUC or a loss, as is standard for that stage, and AutoLR reports no offline metric.

*F2: The selection data are often unstated, and sometimes they are the reported data.* Of the 17 offline-selecting loops, five name a validation split (Self-EvolveRec, RecHarness, VirtualMLE, EvoSkillRec, and Google's validation loss), and Astar ranks candidates with a reward model and accepts by training-loss decrease. Four select on the data they report. RankEvolve-HSTU's persistent leaderboard is the canonical full-test NDCG@10 [23]; Amazon's AutoResearch "retains modifications that improve a held-out scalar metric" and headlines the trajectory's peak [18]; EvoPilot selects and reports on one held-out replay set [35]; and Population Evolution selects on its source instances' evaluation scores and reports its local gains on them, while withholding the transfer targets from the proposer, and calibrates the shared baseline configuration on the released test split [119]. All four disclose these choices, which is to their credit. Seven do not say which data keep or reject candidates; the number that name a non-test selection signal is therefore between 6 and 13 of 17. Outside recommendation, the travel-search co-scientist keeps or reverts each change on the same held-out metric that it reports [159].

*F3: Test-once is rarely stated.* Three offline-selecting loops state that the selected candidate was evaluated on held-out test data once after the search (RecHarness, VirtualMLE, EvoSkillRec); three report on the data they select on (RankEvolve-HSTU, Amazon, EvoPilot); Population Evolution does so for its transfer targets only; ten do not say, so the true number is between 3 and 13.

*F4: No headline result comes with its spread over repeated sessions.* None of the 17 offline-selecting loops repeats an evolution session and reports the spread. Seven have partial evidence: RecHarness and VirtualMLE average three runs without a standard deviation; Google runs six independent sessions with bracketed ranges, but only for an ablation; A-MLE re-runs a candidate when within-run

**Table 4: Evaluation audit (full text, 22 October 2026). Level: evolved object. C1 metric: Full = full-catalogue ranking; Sampled = ranking against sampled negatives; Rank? = ranking metric with the candidate set not stated; AUC = AUC/logloss/loss proxy; A/B = online only; other = a non-recommendation metric; ? = no metric stated. C2 selection signal: val, test (the data that are also reported), loss (training loss), A/B, NS. C4 P: repeats without spread, repeats in an ablation only, a re-run rule, or a noise estimate of single models. C5 P: partial transfer evidence. C6 P: unmatched budget, LLM still proposing, or replay only. C8: Y tuned, N untuned, P partly or on test data, prod. = production model, pub. = published numbers, seed = the loop's seed program. C9 isolation design: P part of the loop constrained (fixed evaluation code, gates, whitelists, sandboxing, bounded actions); $^{G}$ = the paper reports gaming, metric inflation or leakage by the loop. Code: later = promised. Y = yes, N = no, P = partial, NS = not stated, – = not applicable.**

| System | Level | C1 metric | C2 select | C3 test once | C4 repeats | C5 gap | C6 null | C8 baselines | C9 isolation | A/B | Code |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Self-EvolveRec [66] | L2 | Sampled | val | NS | N | P | P | N | NS | N | Y |
| RecHarness [87] | L1+L2 | Sampled | val | Y | P | N | P | NS | P | Y | Y |
| Google/YouTube [147] | L1+L2 | AUC | val | NS | P | P | N | prod. | P | Y | N |
| EvoRec [106] | L1+L2 | Rank? | NS | NS | N | N | P | NS | NS | Y | N |
| VirtualMLE [16] | L1 | Rank? | val | Y | P | P | Y | Y | Y | N | Y |
| NOVA [94] | L1+L2 | AUC | NS | NS | N | N | Y | prod. | P | Y | N |
| AgentX [71] | L1+L2 | A/B | A/B | – | N | P | N | prod. | P | Y | N |
| AgentX-Model [169] | L2 | AUC | NS | NS | P | N | P | prod. | P | Y | N |
| Auto-RecSys [76] | L2 | ? | – | – | N | N | N | prod. | NS | N | N |
| AutoLR [185] | L1+L2 | ? | NS | NS | N | N | N | prod. | P | Y | N |
| CORAL [6] | L1 | A/B | A/B | – | N | N | N | prod. | P | Y | N |
| AgenticRecTune [160] | L1 | A/B | A/B | – | N | N | N | prod. | P | Y | N |
| Sortify [25] | L1 | A/B | A/B | – | N | P | N | prod. | P | Y | N |
| GEARS [175] | L1 | A/B | A/B | – | N | P | N | prod. | P | Y | N |
| RecEvolve [117] | L2 | Rank? | NS | NS | N | N | N | prod. | P $^{G}$ | Y | N |
| Astar [51] | L2 | Rank? | loss | NS | N | P | P | prod. | NS | Y | N |
| A-MLE [39] | L1+L2 | AUC | NS | NS | P | N | P | prod. | P | N | N |
| SR-Agent [167] | L1 | A/B | A/B | – | N | P | N | prod. | P | Y | N |
| EvoSkillRec [78] | L2 | AUC | val | Y | N | Y | Y | P | P | N | Y |
| RankEvolve-HSTU [23] | L1+L2 | Full | test | N | N | P | P | pub. | N $^{G}$ | N | later |
| Population Evolution [119] | L2 | Full | test | P | N | Y | P | P | P | N | later |
| Trajectories-to-Evidence [201] | L2 | AUC | NS | NS | N | P | P | prod. | P | Y | N |
| EvoPilot [35] | L1+L2 | Rank? | test | N | P | P | N | prod. | P $^{G}$ | Y | N |
| Amazon AutoResearch [18] | L1+L2+L3 | Full | test | N | P | P | N | Y | P $^{G}$ | N | N |
| PILOT [82] | L1 | A/B | A/B | – | N | P | N | prod. | P | Y | N |
| A/B Agent [59] | L1 | A/B | A/B | – | N | N | N | prod. | P | Y | N |
| DMRL [188] | L1 | A/B | A/B | – | N | P | N | pub. | P | Y | N |
| *Comparison points outside recommendation* | | | | | | | | | | | |
| EvoRank (LTR) [118] | L2 | other | val | Y | Y | Y | Y | Y | Y | N | Y |
| AI Co-Scientist (search) [159] | L2 | AUC | test | N | N | N | N | P | NS $^{G}$ | N | N |
| RankEvolve-IR (lexical) [112] | L2 | Full | NS | Y | N | Y | P | N | N | N | Y |
| Evolving Semantic Indexes (GR) [111] | L3 | Full | val | Y | P | Y | N | seed | Y | N | N |
| AgentLTV (LTV) [156] | L2 | other | test | NS | N | N | P | NS | N | N | Y |
| Feedback Contract (CTR/LTR) [171] | L2 | AUC | val | Y | Y | Y | N | seed | Y | N | N |
| PEAR (search) [150] | L1 | A/B | A/B | – | P | P | N | prod. | NS | Y | N |

variance is high, which the authors say "materially reduced" single-seed promotions [39]; and AgentX-Model, EvoPilot and Amazon estimate the noise of single models without repeating a session—EvoPilot notes that two publications of the same model, configuration and pinned data "differed by 0.36 points" [35], and Amazon re-evaluates one model across 11 dates ("6.18% ± 0.28 pp") [18]. Several papers state that their gains are within the resolution of a single run: EvoSkillRec notes margins "on the order of $10^{-4}$ AUC, which a single evolution run cannot reliably separate from run-to-run variation" [78]; AgentX-Model reports run-to-run standard deviations of 0.0010–0.0022 AUC while counting an experiment as above baseline if it exceeds it by more than $10^{-6}$ [169]; and AutoLR, in its audited configuration, promoted candidates after a single run with noise calibration disabled, a risk its authors name the "KEEP ratchet": each promoted model becomes the baseline for all later comparisons [185]. None of the nine online-selecting loops repeats a session either; for them a repeat is only practical on a replay (R4 in §7).

*F5: Where transfer is checked, some selected gains survive and some do not.* Two core loops report an explicit held-out check of

their selected candidates. In EvoSkillRec, five candidates lie on the validation Pareto front, "but only P1 (round 19) retains both improvements on the test set" [78]; P1 has the highest validation score of the five, so choosing by validation would have picked it (the paper states no rule for choosing a front point); P2 and P3 lose their AUC gain on test, and P4 and P5 have none on validation. In Population Evolution, a ranking candidate gains 10.66% on a peer dataset and loses 0.79% at larger scale, two of three peer-screened designs keep their gains, and sharing evidence across searches raises the larger-scale pass rate for ranking from 11/60 to 18/60 [119]: in ranking and RL, most locally selected candidates do not graduate. Two further observations point the same way. Amazon's headline lift of 1.82× is the peak of a trajectory, "~2.6$\sigma$ above the across-date mean" of 1.63× [18], and Kuaishou's trajectory analysis finds that for 22 of 30 methods the last round falls below an earlier best [201]. The comparison points give larger effects: in EvoRank, the gains of the first (loss-evolution) campaign "were almost entirely fitness noise" on 60k held-out queries, while about 70% of the second campaign's gains survived [118]; in an e-commerce search loop, effects on a shadow-traffic proxy were several to tens of times larger than in decision-grade online experiments, and five promoted candidates showed no significant gain [150]; and in Evolving Semantic Indexes, a proxy decoder that ranks semantic-ID programs well (Spearman $\rho = 0.90$) becomes "not only less informative but misleading" once the training objective may change, on a small sample of five programs [111]. A controlled study shows that the mechanism exists: on a label-permuted task with no real signal, a coding agent given exact validation scores "ended by reporting a gain of 0.0157" [171]. None of this measures how much of the published recommender gains is selection noise; it shows that the question cannot be settled without held-out checks.

*F6: Null controls are rare.* Three offline-selecting loops compare against a non-LLM search at an equal or favourable budget: VirtualMLE (grid search and Bayesian optimisation), NOVA (Optuna-TPE at a matched offline-evaluation budget, in its code-level setting) and EvoSkillRec (NASRec, alongside OpenEvolve); in all three the LLM loop is reported to win. Nine have partial controls (unmatched budgets, a random arm router that still uses LLM edits, cross-LLM swaps, or replay-only comparisons), and five have none. No loop runs an equal-budget random search over its own edit space, and none runs the catalogue-of-known-techniques null of §5.8. Where random selection was run, it was competitive: in AgentX-Model's historical replay, which randomises the choice of experiments rather than the edits, in Scenario A "Uniform Random's mean exceeds Fixed, Greedy, and Joint" policies [169], and in EvoRank's loss campaign an equal-budget random search transferred as well or better ($p = 0.03$) [118]. AutoLR states plainly that it "does not establish superiority over random, greedy, or direct LLM-based routing under a matched trial budget" [185]. The history of architecture search explains why the control matters: random search with early stopping matched leading NAS methods [75], the search phase of several methods was no better than random sampling [166, 173], and classical optimisers beat LLM agents in fixed hyperparameter spaces [34, 128]. The online-selecting loops compare each candidate with a control arm, which is a different question from whether the proposer beats a null search.

**Table 5: Shortcuts reported in the audited loops (F7): what happened, what caught it, and a guard that prevents it by construction.**

| Failure | Where | Caught by | Guard |
|---|---|---|---|
| Metric definition changed (batch size inflates an in-batch metric) | RecEvolve | a human | freeze metric-relevant settings |
| Target leakage in features or training pairs | RankEvolve-HSTU | the agent, from anomalous leaderboard trajectories | leakage checks before search (NOVA, EvoRank) |
| Evaluation protocol changed (multi-position scoring) | RankEvolve-HSTU | mandatory diagnosis of negatives | evaluator the agent cannot edit |
| Evaluation date inside the training window | EvoPilot | not stated | date-order checks |
| Proxy metric gamed (codebook size) | Amazon AutoResearch | human QA between runs | a second, usability metric |
| Capacity silently increased | AI Co-Scientist | not stated | matched-capacity comparisons |

*F7: Agents find shortcuts in the evaluation, and the keep-or-reject rule does not catch them.* Four core loops and one comparison point report metric-winning bugs, metric inflation or leakage produced by the loop itself (Table 5). RecEvolve's agent "discovered that drastically reducing the batch size (from 8k down to 1k) artificially inflated proxy metrics"; a human caught it and rolled it back [117]. On HSTU, RankEvolve-HSTU's agents produced a genre feature that leaked the target (training loss collapsed to zero while evaluation spiked), DPO preference pairs mined at the held-out target position, and a naive multi-position evaluation that would have inflated NDCG@10 from 0.2192 to 0.2517/0.2577; the leaks surfaced as anomalous trajectories on the leaderboard and the inflation through a mandatory diagnosis step, none was reported as a gain, but the best agent pair still had a 10.4% silent critical-defect rate on ExecML-HSTU, and a single Claude Code pass 35.4% [23]. In EvoPilot, "a favorable result used an evaluation date inside the candidate model's training window, so the result was retracted" [35]. In Amazon's semantic-ID loop, "maximizing codebook size trivially improves weighted coherence", and the resulting hierarchy was unusable; human QA between runs caught it [18]. Where a paper says what caught the problem, it was something outside the keep-or-reject rule: a person (RecEvolve, Amazon) or the agent's own diagnosis of anomalous results (RankEvolve-HSTU). Coding agents that can reach their tests behave the same way outside recommendation [7, 105, 195]. These four papers should be credited for reporting incidents (R8); a paper that reports none has not shown that none occurred. Among the borderline systems, AURA's agent fabricated 904 session identifiers in its diagnostic stage [65], and RecSys Factory's diagnostics uncovered a pre-existing upstream join bug that had "silently corrupted all downstream training samples" [5], a reminder that the data feeding the loop need checks too. As designed, only VirtualMLE states that the test set is hidden from the agent and the data pipeline immutable; twelve offline-selecting loops constrain part of the loop (verification gates, fixed evaluation code, whitelists, isolated workspaces, sandboxed modules),

RankEvolve-HSTU lets its agents run evaluations at the test position, and three say nothing; all nine online-selecting loops bound the agent's actions.

*F8: Baselines, budgets and openness.* The largest relative gains are measured against weak starting points. RecHarness improves average HR@10 by 85.85% over a cold-start GRU4Rec [49] template and by 12.58% over HSTU [87], and Self-EvolveRec configures all models with the same fixed hyperparameters (embedding 50, batch 128, learning rate 0.001) without describing tuning them [66]. Against production baselines, online lifts on broad metrics such as watch time, revenue or GMV are mostly between 0.06% and 5% (`design.csv`). Only VirtualMLE and Amazon's AutoResearch (a hand-tuned baseline from four engineer-weeks) describe tuning their baselines on non-test data; Population Evolution tunes them on the test split. Budgets are reported in incompatible units (iterations, GPU-seconds, GPU-hours, ideas, tokens, dollars, yuan); twelve of 27 report LLM tokens or cost. Four of 27 release code, two promise to, and six do not name the LLM they use.

*F9: Online gates are reported, but rarely as confirmations.* Nineteen of the 27 loops report online experiments; these are the main safeguard of the industrial loops and strong evidence when well designed. Nine state the experiment duration and nine report significance or confidence intervals; none corrects for multiple testing across the many rounds a loop runs (AutoLR discusses the problem but applies no correction), although the arm chosen as best overstates its effect [11, 72]. AutoLR's summed launch-review lifts are, in its own words, "descriptive and are not pooled treatment effects" [185]. Only Sortify re-tests its chosen round in a fresh seven-day experiment [25]. PILOT designs its final confirmation as "a single pre-registered test" that "runs once", because "the data collected while searching … cannot also serve as proof" [82], but the report does not yet include that confirmation, and its headline compares the best of five buckets before and after, "not randomized". Online noise is visible inside single papers: DMRL's default configuration (Qwen3-8B, eight rollouts) appears in six tables run in "non-overlapping time windows" with LTV lifts from +0.021% to +0.064%, a spread of 0.043 percentage points, larger than its 0.028-point lead over the best baseline in the main table; the authors restrict their conclusions to within-table comparisons [188].

*Practices worth copying.* The audit also found practices that, if combined, would meet most of the standard in §7: RecHarness's validation-only selection with a single test evaluation; VirtualMLE's hidden test set, immutable pipeline and grid-search and BO baselines; EvoSkillRec's validation-versus-test check of its Pareto front; Population Evolution's withheld transfer targets; NOVA's leakage verifier and EvoRank's pre-search leakage audit with an equal-budget random search; RankEvolve-HSTU's mandatory diagnosis of negative results; and Sortify's fresh re-test and PILOT's pre-registered confirmation design.

*Does selection really overfit?* A meta-analysis of 120 Kaggle competitions found little overfitting to public leaderboards [129], and new test sets for ImageNet showed accuracy drops but preserved model rankings [124], partly because similar models overfit a reused test set less than theory predicts [100]. Agent-proposed candidates are near-copies of an incumbent, which by that argument limits regressional overfitting. Two differences remain: an agent loop evaluates hundreds of correlated candidates on one split without a resubmission cap [13] and keeps the maximum, and it controls the code that produces the score, which no Kaggle team does. The direct measurements (F5) show that some selected gains shrink or vanish on transfer while others survive. The safe conclusion is not that self-evolving recommenders do not work, but that the published evidence cannot yet tell how much of their gain is real.

## 7 A Reporting Standard

The audit suggests that most of the problems are cheap to report: they are unreported or unenforced rather than unsolvable. Table 6 turns the codebook into a reporting standard. Each item has a minimum that any paper can meet by reporting what it already did, and a recommended practice that costs compute; each maps to the audit criterion that checks it, so that a paper's compliance can be read off the same coding. The items extend general checklists for NAS research [86], ML-based science [62] and agent benchmarks [200] to the specific failure points of an evolution loop (Figure 1). Appendix A applies the minimum tier to two audited papers.

*Cost of compliance.* The minimum tier costs nothing beyond writing. In the recommended tier, R4 and R5 multiply training cost: three sessions plus one equal-budget null cost four sessions. For an academic harness such as RecHarness, whose budget is 43,200 GPU-seconds per backbone [87], that is 48 GPU-hours per backbone. LLM costs are small next to training: VirtualMLE reports $1.5–4.8 of LLM cost per session [16], EvoRank about ten dollars for fifty iterations [118], and NOVA about $29 per task [94] (LLM cost only). For industrial loops whose candidates take hundreds of GPU-hours [76], R4 and R5 can be applied to a replay or a reduced-scale proxy, as AgentX-Model does for scheduling policies [169], while R9 governs the launch decision.

*A shared protocol.* Several items can be enforced in software rather than requested in prose: a sealed test split (R2, R3), a single final evaluation, an equal-budget null evolver (R5), a budget card (R6) and an evaluator outside the agent's reach (R8). Appendix B describes recforge [123], a reference implementation that we are building; it enforces R2, R3 and R5 through its API but not yet R8, and we report no results from it.

## 8 Open Problems

*Isolation without crippling the agent.* Code-level agents need to read data and change training code, and that is exactly the access that lets them leak targets or change what the metric measures (F7). Verification gates (NOVA), fixed evaluation code (AgentX-Model), immutable pipelines (VirtualMLE) and cross-agent review (RankEvolve-HSTU) each close part of the gap. A principled design would give the agent a process that physically cannot reach evaluation labels or code, and would audit every metric-relevant change, such as batch size for in-batch metrics.

*Proxies for expensive candidates.* Industrial candidates take hours to hundreds of GPU-hours, so loops screen on proxies: small decoders [111], data subsets and peer datasets [119], training-loss curves and learned reward models [51], and LLMs can themselves predict a candidate architecture's performance before training [57].

**Table 6: Reporting standard for self-evolving recommenders. *Minimum*: report what was done; *recommended*: what a strong study does. The last column gives the audit criterion (Table 1) or finding that checks the item.**

| | Minimum (report) | Recommended (do) | Audit |
|---|---|---|---|
| R1 Metric | State the candidate set and how negatives are drawn (full catalogue, $N$ sampled, in-batch); for ranking-stage models, the traffic on which AUC or logloss is computed. | Full-catalogue ranking where the catalogue allows, sampled metrics only as secondary [69]. | C1 |
| R2 Selection | Name the data or signal that accepts candidates at every stage of the chain (proxy, offline, online). | Keep the test split out of every decision, including baseline tuning. | C2 |
| R3 Test once | Report the selected candidate's validation score next to its test score. | Evaluate on test once, after the search. | C3, C5 |
| R4 Sessions | Run $\geq 3$ independent sessions and report every session's final test score. | $\geq 5$ sessions with an interquartile mean and bootstrap interval [1], and the expected best score as a function of budget [29]. Industrial loops: repeat on a replay or a reduced-scale proxy. | C4 |
| R5 Null control | One null at equal trial and compute budget: HPO or random search over the knobs that the kept edits changed. | A ladder of nulls: (i) HPO over the same knobs, (ii) random mutation over the edit arms or grammar, (iii) the same proposer without evaluation feedback or with candidates accepted at random. | C6 |
| R6 Budget | A budget card: trials, GPU-hours, LLM tokens and cost, wall-clock, human hours, and the cost of training a learned proposer. | Report results as a function of budget, not only at its end [64]. | C7 |
| R7 Baselines | How the starting point and compared models were tuned, and on which data. | Tune them with the budget of one session. | C8 |
| R8 Access | What the agent can read and write (data, labels, evaluation code, metric definitions); every gaming or leakage incident and what caught it. | An evaluator in a process the agent cannot modify; review of metric-relevant changes (batch size, candidate set, evaluation date). | C9 |
| R9 Online gates | Duration, traffic share, pre-specified primary metric and guardrails, confidence intervals, and the number of rounds tested. | A correction across rounds and a fresh, pre-registered confirmation of the final candidate, reporting the confirmed rather than the selected effect [68]. | F9 |

The evidence shows that proxy fidelity depends on what is evolved. Characterising when a proxy can be trusted, and spending budget to re-check proxies during search, is open.

*Budget-normalised comparison of evolvers.* There is no common unit of cost and no common task, so the field cannot say whether bandit routing, island evolution, research-agent pairs or learned proposers are better. A shared benchmark with a budget card (R6) and an equal-budget random baseline (R5) would allow the first such comparison.

*Evolving generative recommenders.* Semantic-ID construction, codebook collisions, constrained decoding and preference-alignment losses are new design dimensions with little human prior and large effects. They are natural targets for code-level evolution, and almost untouched (§5).

*From offline wins to online value.* RecEvolve observes that offline evaluations often show "win-win" results while online tests show trade-offs [117]. Loops that select on offline metrics inherit the gap; loops that select on online experiments face multiple testing and slow iteration. Combining the two with a fresh online confirmation of the final candidate, as Sortify does with a seven-day re-test, ideally pre-registered, is a starting point.

*Circularity at the policy level.* L4 systems are often judged by simulators from the same model family as the agent being improved. Held-out simulators [66], verifiable rewards [108] and reward-model benchmarks [176] are partial answers.

## 9 Threats to Validity

*Coverage.* The field is months old and grows weekly; our cut-off is 22 October 2026. Many systems are arXiv preprints whose versions change; we coded the version available on that date. Generic keyword search performed poorly, so coverage relies on citation-graph search and may miss systems that neither cite nor are cited by the core set.

*Coding.* Each cell was drafted from the full text with verbatim evidence and then checked against the original papers in two adversarial verification passes, run by LLM agents and reviewed by the authors: the first corrected 12 cells in nine rows, the second re-verified every cited sentence against its source and corrected the remaining cells and claims (both reports are released in `review/`). A third pass, prompted by an internal review of this version, recoded nine criterion cells and five object labels, moved meta-evolution into its own column, and separated C9 into isolation design and reported incidents (listed in the released `audit/README.md`). All of this is one reading. A second author is independently coding all 27 core systems from the full texts without access to the first coding; we will report, before any reconciliation, the raw agreement and Cohen's $\kappa$ per criterion with bootstrap intervals, a prevalence-robust coefficient (Gwet's AC1), and the confusion matrices, and we will flag in the text every count that rests on a criterion with $\kappa < 0.6$. "Not stated" does not mean "not done": authors may have followed a practice without reporting it, which is why we give ranges for the counts that rest on such cells. We have not yet asked the audited authors to check their rows; the evidence is released so that they

and other readers can correct individual cells, and we will update the table.

*Industrial systems.* Offline hygiene criteria mean little for loops whose keep-or-reject decision is an online experiment. We therefore report those loops separately and assess them on the design of their experiments (F9), and we do not treat a missing offline test split as a defect when no offline claim is made.

*Conflict of interest.* The authors are building recforge, the reference implementation described in Appendix B; the audit does not depend on it. One of us wrote the ideation-harness study cited in §5, which is outside recommendation and not peer reviewed.

## 10 Conclusion

Self-evolving recommenders moved in less than a year from an idea to production systems at many large platforms. The loops are inventive, and several report online gains under well-designed experiments. The offline evidence is weaker than the headlines suggest: selection data are often unstated, test-once and repeated sessions are rarely reported, equal-budget null searches are rare, and the few papers that check transfer find that some selected gains do not survive. Online gates are reported but seldom corrected for the rounds they test or confirmed afresh. Agents also find shortcuts in the evaluation, which people or the agents' own diagnoses, not the acceptance rules, catch. Most of this is cheap to report. We hope that the reporting standard in Table 6, the released audit and design data, and shared protocols that enforce the standard in software make it the norm, so that the next comparison of evolvers can be made on equal terms.

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## A The Minimum Tier Applied to Two Papers

Table 7 applies the minimum tier of Table 6 to an open recommender harness (RecHarness) and to the comparison point that meets the most criteria (EvoRank), using only the audit evidence. It shows that the standard can be checked from a paper's text, and that a study can meet nearly all of it at a cost of about ten dollars of LLM calls per run.

**Table 7: Minimum-tier compliance of two audited papers, from the released evidence. Y met, P partly met, N not met, – not applicable.**

| | RecHarness | EvoRank | Note |
|---|---|---|---|
| R1 | Y | Y | 99 sampled negatives stated; per-query full lists |
| R2 | Y | Y | both select on a validation split |
| R3 | P | Y | RecHarness tests once but reports no validation score beside it |
| R4 | P | Y | three runs in both; RecHarness reports only their mean |
| R5 | P | Y | RecHarness's random arm router keeps the LLM edits; EvoRank runs equal-budget random search |
| R6 | P | P | GPU-time budget, but the LLM is not named; EvoRank gives dollars but not the LLM |
| R7 | N | Y | cold-start template vs. Optuna-tuned LambdaMART |
| R8 | P | Y | isolated workspaces vs. leakage-guarded helpers and a pre-search audit |
| R9 | P | – | 7-day A/B on 10% of traffic; significance not stated |

## B recforge: A Reference Implementation

recforge [123] is an open benchmark that we are building to enforce the parts of the standard that software can enforce. It is under development, it has produced no evolver results, and the audit does not depend on it. Its design follows the audit (Figure 4).

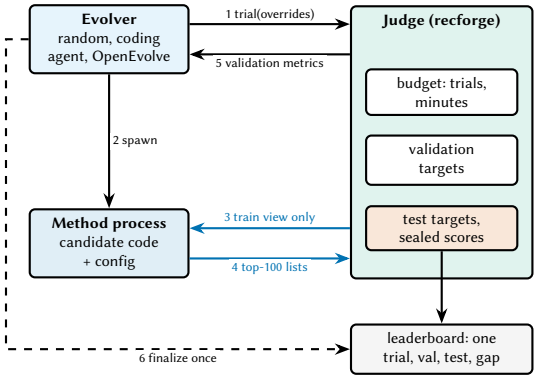



**Figure 4: An evolve session in recforge. The evolver requests trials (1); each trial runs the candidate in its own process (2), which receives a training view without targets (3) and returns ranked lists (4). Only validation metrics flow back (5); test scores are computed and sealed. One `finalize` publishes a single trial with its validation score, test score, gap and budget (6). The seal holds at the API level; file-system isolation is planned.**

- *One protocol per version.* The current protocol is leave-one-out per user (last item test, second-to-last validation) with full-catalogue ranking, the user's history removed, and HR and NDCG at 10 and 20 and MRR@100 (items ranked below 100 score zero). Leave-one-out ignores the global timeline and can leak future information [48, 58]; a global temporal split is the next protocol version. Results from different versions are never put in one table, and every result is keyed by the source hash of the method, its configuration and the code commit.
- *A sealed test at the API level.* Method code runs in its own process and receives only a training view; asking for a validation or test target raises an error, and the method returns ranked lists that the parent process scores. The seal holds only against code that uses the API: sealed scores and session state are ordinary files, so an agent with file-system access could read test scores or alter the budget. A container mode that would close this gap (R8) is planned, not implemented.
- *Budgeted evolve sessions.* A session has a trial and time budget. Each trial returns validation metrics only; test metrics are computed and sealed. A single `finalize` reveals the test metrics of exactly one trial, publishes it with the session's budget, and closes the session.
- *Evolvers on equal terms.* An equal-budget random search (the R5 null), a wrapper for an external coding agent, and an OpenEvolve adapter drive sessions through the same interface, and each run reports the validation-to-test gap of its finalized trial. They are tested on synthetic data only; the OpenEvolve adapter has not been run end to end.
- *Baselines and data.* Popularity, ItemKNN [131], BPR [126], SASRec [61], HSTU [177] and TIGER [122], with MovieLens-1M [44], Amazon 2014 Beauty, Sports and Toys [102], MIND news [157], and the OpenOneRec release [196].

Not yet implemented: the container mode, recording of LLM tokens and cost, automatic aggregation of repeated sessions, and the temporal split.